

EMSJUMP: Electrical Muscle Stimulation as a Wearable Training Tool for Take-off Phase of Ski Jumping

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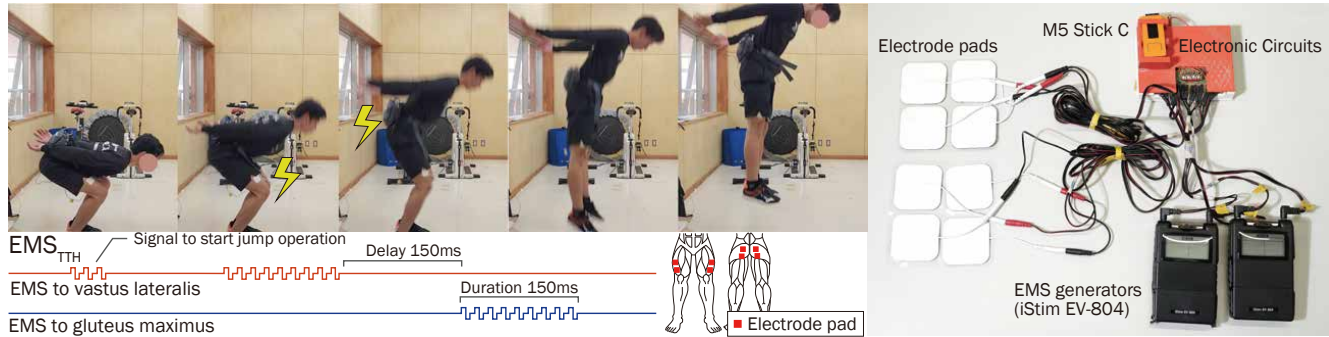


Figure 1: Left: The EMS_{TTH} pattern and sticking positions of electrode pads. View of the user study. Right: our system.

ABSTRACT

We developed a wearable Electrical Muscle Stimulation (EMS) system aimed at enhancing kinesthesia and promoting skill acquisition in ski jumping. Through experiments, it was demonstrated that the application of EMS during jumping significantly increased the angular velocity of knee extension and altered perception.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

electrical muscle stimulation, kinesthesia, motor skills learning

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1 INTRODUCTION

Mastering ski jumping's take-off movement is intricate, demanding quick, coordinated action within only 0.3 seconds, requiring precise

bodily awareness and skill. Biomechanical analysis has established a connection between the length of the jump to factors such as inrun velocity, force at take-off [Virmavirta and Komi 1993], and air resistance when in flight [Virmavirta et al. 2005].

Despite advances in biomechanical simulations and analyses, the understanding of the origins and conscious experience of movement in sport remains limited, including the tacit knowledge acquired through direct engagement with movement [Åsa Bäckström 2014]. The mastery of movement encompasses more than just observation of movement through visual and auditory feedback; it requires a deep, internal kinesthetic awareness emanating from our own corporeal movement [Sheets-Johnstone 2010].

To address this educational void, we introduce a wearable Electrical Muscle Stimulation (EMS) training device, which induces involuntary muscle contractions at intended timings in the muscles of the lower limbs. This enables training to better understand how to use the muscles during ski jumping take-off movement.

2 SYSTEM

Our system is a prototype system that enhances movement by applying EMS to lower limbs during jumping movement (Figure 1). The EMS can be applied at intended timings and for any duration. Electrode pads were applied to the skin where the gluteus maximus and vastus lateralis, the main extensor muscles of lower limbs, were located. Operation is possible externally via Bluetooth, and the system can be stored in a waist pouch (total weight: 1151 g). EMS applied to the buttocks results in backward motion of at least 10 cm in the raised foot's toes, while EMS to the thighs causes full knee extension. We have fine-tuned the stimulation timing to account for the natural reaction time during a jump. We establish

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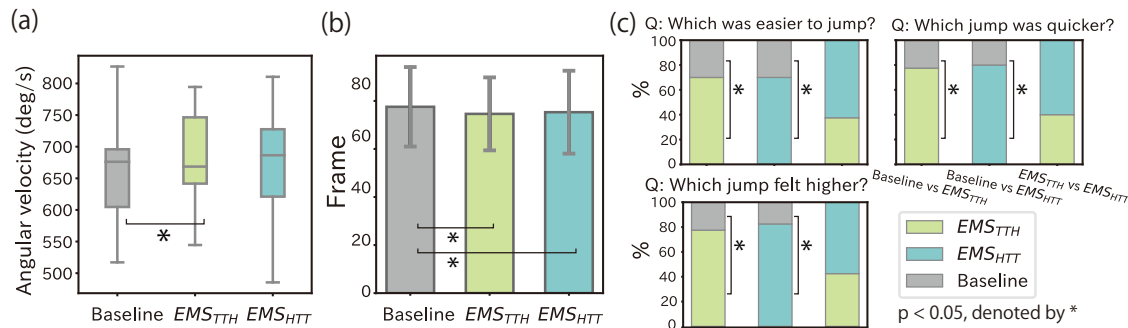


Figure 2: (a): Peak angular velocity of the knee during squat jumps. (b): Time from resting position to peak knee velocity. (repeated-measures ANOVA) (c): Comparative results of sensory perception of two different EMS patterns (binomial test).

three experimental conditions: *EMS_{TTH}* (Thigh to Hip), with stimulus originating in the vastus lateralis and then proceeding to the gluteus maximus; the reverse order, *EMS_{HHT}* (Hip to Thigh); and Baseline (without EMS) condition. *EMS_{TTH}* is based on a study that EMG in a ski jumper activated gluteus maximus after about 150 ms in vastus lateralis [Virmavirta 2001]. The duration of the EMS was determined by empirical experiments that determined the magnitude of involuntary movements and how the stimulus was felt.

3 FOUNDATIONAL EVALUATION

Since the effects of patterned EMS during jumping movements have not been examined, a basic study was conducted to examine the effects of physically facilitating jumping and the cognitive changes of jumping in volunteers (N=20) who perform vertical jumping. The jumps were captured using a motion capture system (Optitrack, Acuity Inc., Japan) that operated at a frequency of 250 Hz. After each jump, participants were repeated three questions comparing two different EMS patterns: "Which jump was easier to perform?", "Which jump did you feel that you reached a higher point?", and "Which jump was executed quicker?".

Our analysis of the resultant data indicated that the *EMS_{TTH}* intervention enhanced the peak angular velocity of the knee during squat jumps and both EMS patterns accelerate the time from a resting position to that peak velocity (Figure 2). Participants also noted discernible changes in their sensory perceptions, citing that jumps conducted with EMS assistance felt more effortless, achieved greater heights, and were executed more quickly compared to jumps without EMS. These findings reinforce that EMS applications can substantially alter performance and a user's perceived exertion during jumping activities.

4 USER STUDY

Our user study involved four ski jumpers (age 16 - 18 years; four males). The participants, all of whom have competed at national competitions for high school students, included one individual who placed fourth at a national level and another who earned second place in the Junior Olympics. The participants engaged in popular dry land training exercises designed to imitate take-off movement

during which they also experienced our EMS system. As part of the study, they verbally stated their kinesthetic sensations triggered by the EMS, responding to semi-structured interview questions.

The study results showed that participants responded positively to the *EMS_{HHT}* pattern, reported a distinct sensation of initiating movement from the hips. This pattern was also reported to facilitate more effective recruitment of the gluteus maximus, which resulted in prolonged periods of muscle exertion and heightened sensations of achieving greater jump heights. In contrast, the *EMS_{TTH}* was reported negative comments, as participants commonly observed difficulties in transferring force. One participant particularly struggled with self-directed learning of "moving from hips", he expressed a belief that the concurrent training with EMS could potentially accelerate the learning of muscle coordination and control. These observations imply that EMS could assist individuals, especially those who struggle with learning quick, precise movement, in rapidly developing an implicit understanding of movement that is not visible to others.

5 CONCLUSION AND FUTURE WORK

The study identified two key findings: EMS altered the perception and performance of joint angular velocity during jumping movement. We also found its influence on kinesthesia and its potential for ski jumping skill acquisition. Future work will be conducted in collaboration with athletes to explore and classify the kinesthetic sensations associated with various EMS patterns.

REFERENCES

- Maxine Sheets-Johnstone. 2010. Kinesthetic experience: understanding movement inside and out. *Body, Movement and Dance in Psychotherapy* 5, 2 (2010), 111–127. <https://doi.org/10.1080/17432979.2010.496221>
- M. Virmavirta. 2001. Plantar pressure and EMG activity of simulated and actual ski jumping take-off. *Scandinavian Journal of Medicine & Science in Sports* 11, 5 (10 2001), 310–314. <https://doi.org/10.1034/j.1600-0838.2001.110510.x>
- Mikko Virmavirta, Juha Isolehto, Paavo Komi, Gert-Peter Brüggemann, Erich Müller, and Hermann Schwameder. 2005. Characteristics of the early flight phase in the Olympic ski jumping competition. *Journal of Biomechanics* 38, 11 (2005), 2157–2163. <https://doi.org/10.1016/j.jbiomech.2004.10.004>
- M. Virmavirta and P. V. Komi. 1993. Measurement of take-off forces in ski jumping. *Scandinavian Journal of Medicine & Science in Sports* 3, 4 (1993), 229–236. <https://doi.org/10.1111/j.1600-0838.1993.tb00387.x>
- Åsa Bäckström. 2014. Knowing and teaching kinaesthetic experience in skateboarding: an example of sensory emplacement. *Sport, Education and Society* 19, 6 (2014), 752–772. <https://doi.org/10.1080/13573322.2012.713861>